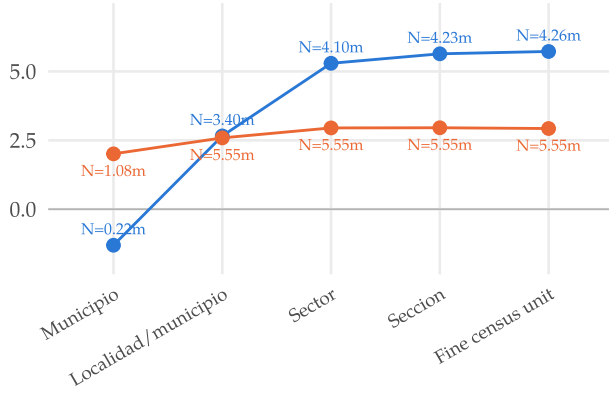


PM10: native Design B gap at 3 km and 20 km, three education groups

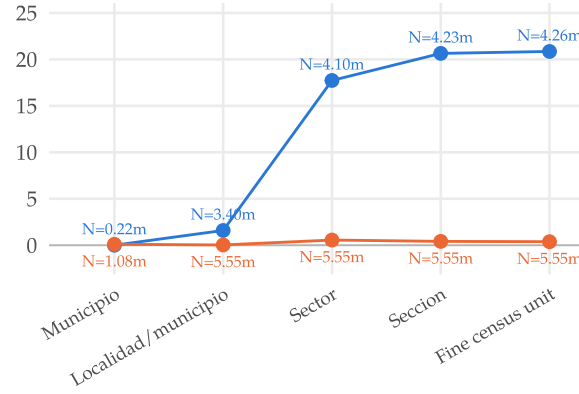
IDW buffer — 3 km — 20 km

Below secondary - Bachelor's+ (outcome units)

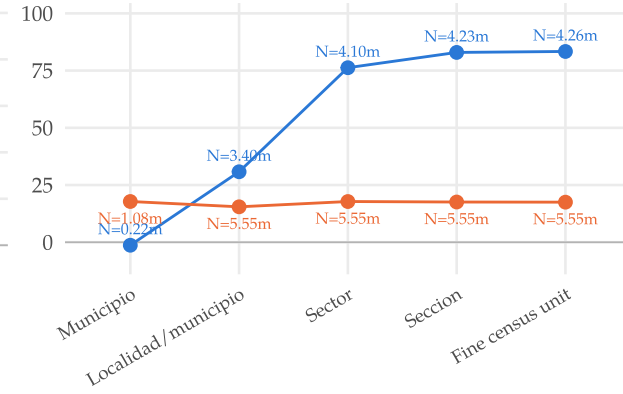
Bogota: annual mean



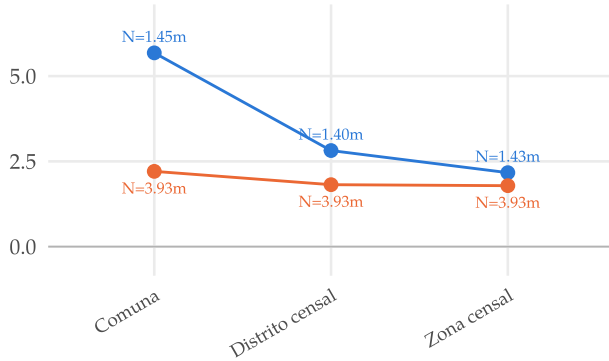
Bogota: hours >= IT1



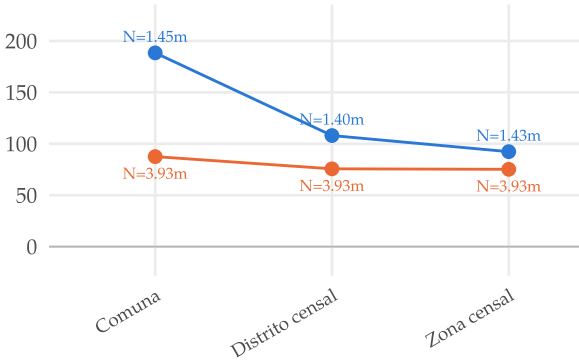
Bogota: hours >= IT2



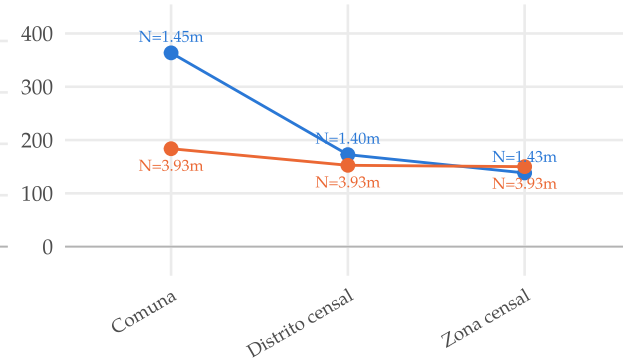
Santiago: annual mean



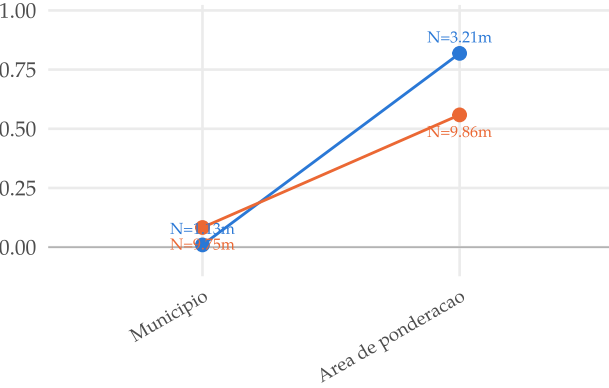
Santiago: hours >= IT1



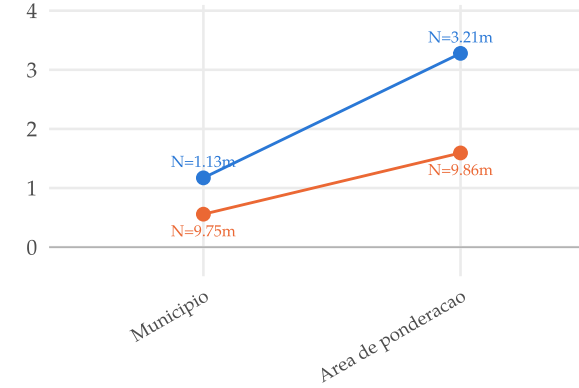
Santiago: hours >= IT2



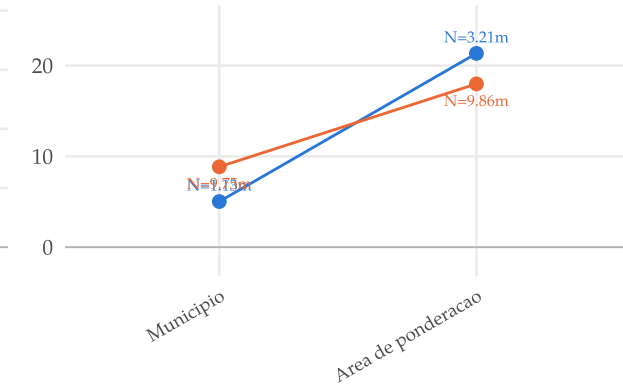
Sao Paulo: annual mean



Sao Paulo: hours >= IT1



Sao Paulo: hours >= IT2



Geographic support (coarse to fine)

Native Design B: exposure rebuilt at each support. Point labels (N) are sample sizes, not gaps: millions of education-reporting adults in covered units. Avg: concentration units; IT1/IT2: hours at or above the threshold in 2023.